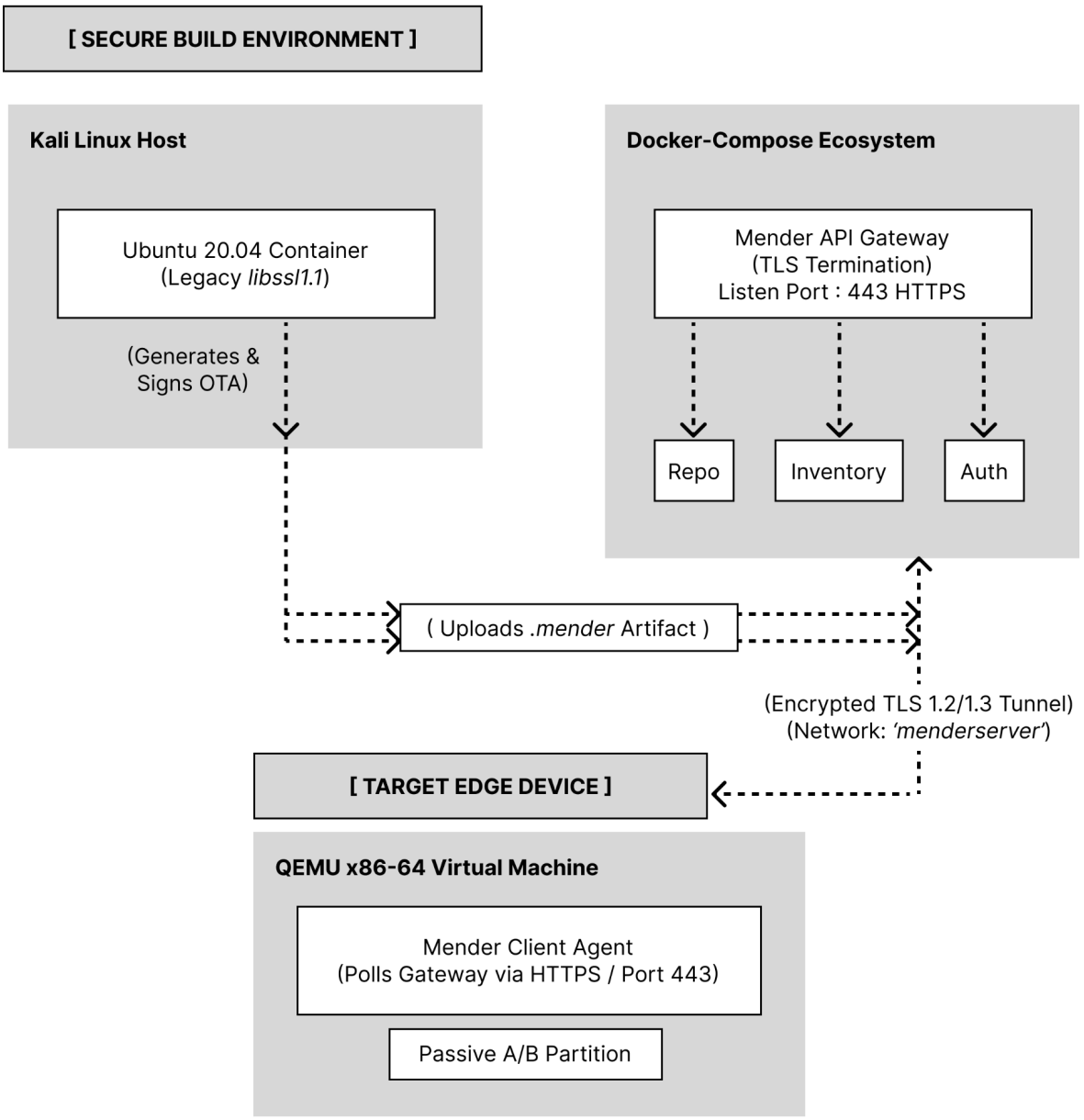
Deliverable 4: Secure Update Server Setup (Task 4)

1. Objective

To deploy and configure a centralized, secure update distribution server utilizing encrypted protocols (HTTPS/TLS) and ensuring authenticated access for authorized IoT devices.

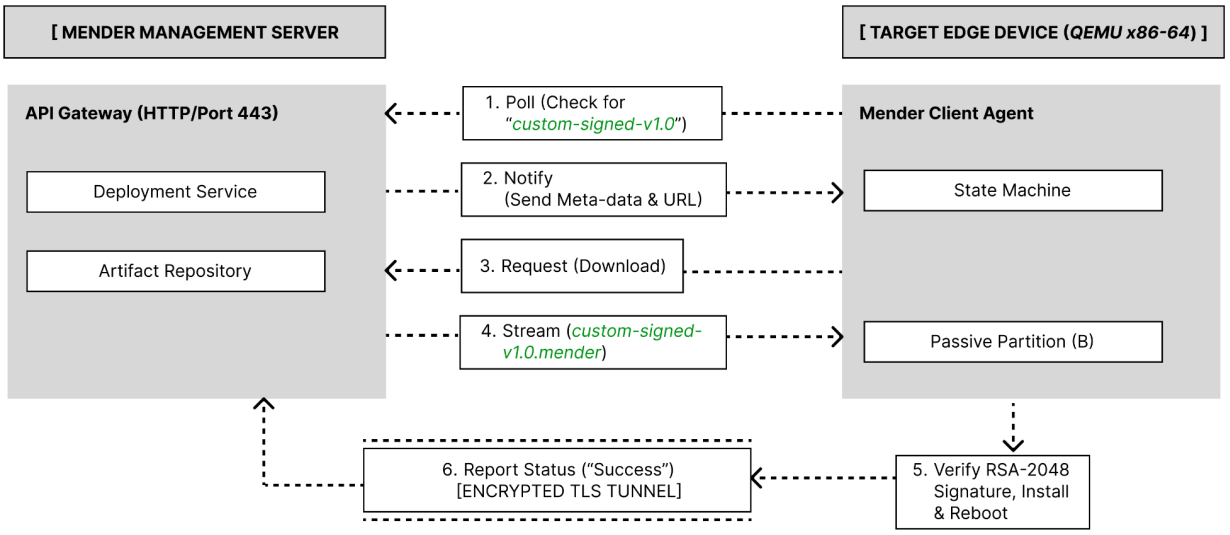
2. Infrastructure Architecture

This blueprint details the secure boundary between your Kali build environment and the Mender management server.



3. Communication & Distribution Flow

This illustrates the secure delivery path for the `custom-signed-v1.0` release to the virtual fleet.



4. Technical Implementation Summary

- **Protocol:** All communications are wrapped in **TLS/SSL**, preventing man-in-the-middle (MITM) attacks during firmware transit.
- **Server Authentication:** The Mender server validates device identities before granting access to specific software releases, ensuring only commissioned devices receive updates.
- **Delta Support:** While currently utilizing full script modules for `custom-v2.0` and `custom-signed-v1.0`, the infrastructure is capable of **Binary Delta** updates to minimize bandwidth on constrained networks.